

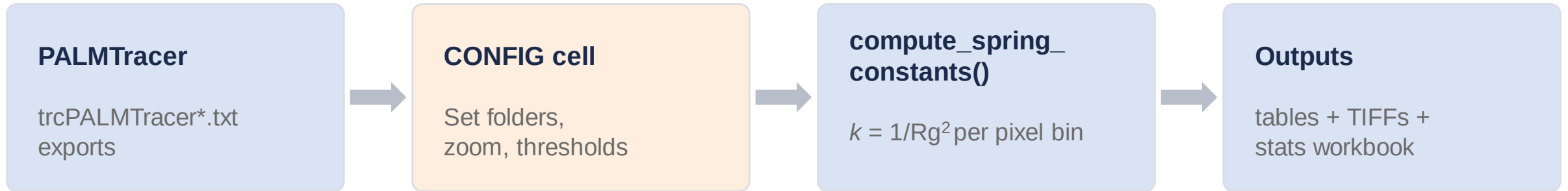
TPM Analysis Manual

Trapping Potential Mapping: Single-Particle Tracking Pipeline

4-TPM_analysis/TPM_Analysis.ipynb

github.com/vanika23/trapping-potential-map-spt

OVERVIEW

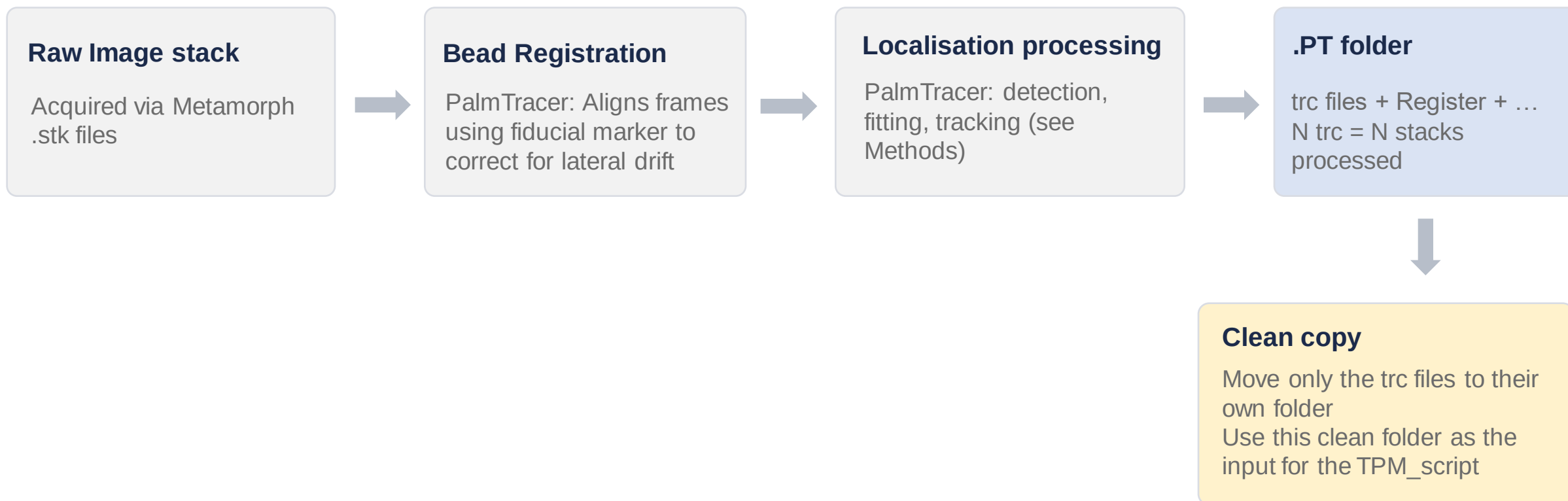


This manual walks the second, third and fourth boxes: setting CONFIG, running the notebook, and reading what comes out.

Note: the pipeline is a Jupyter notebook, not a desktop app, there are no toolbar icons to click, only one CONFIG cell to edit and a Run-all.

BEFORE THE NOTEBOOK

Where trc files come from: Metamorph and PalmTracer



Why copy the trc files out?

The .PT folder keeps accumulating other PALMTracer Output (drift correction, registration logs, state files) as you reprocess. A folder containing only the trc files is a stable, frozen target for the notebook's INPUT_FOLDERS path.

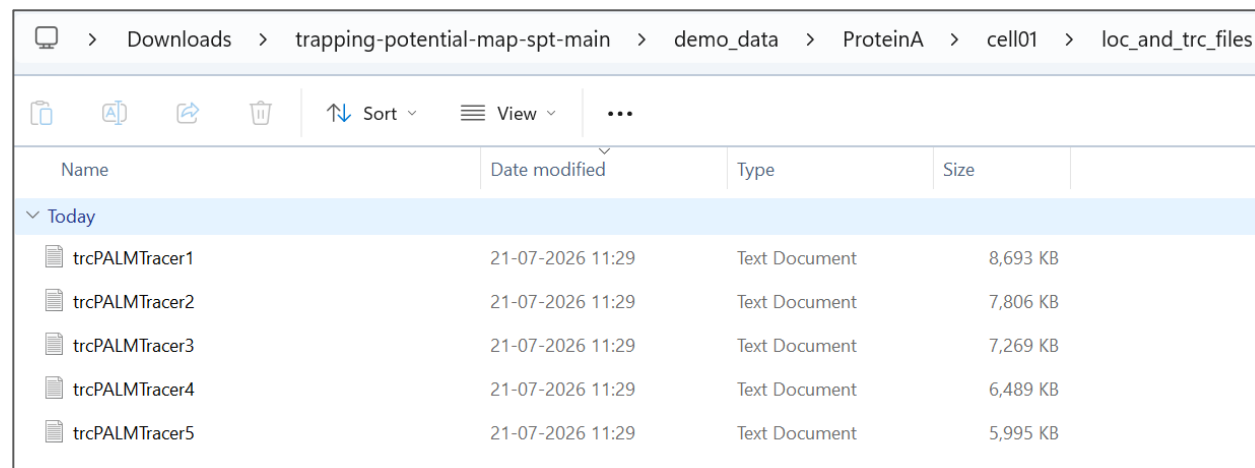
Single-label mode

INPUT_FOLDERS

DATA LAYOUT

Copy trc files out of the .PT folder into their own folder (loc_and_trc_files) following the hierarchy below

```
demo_data/  
|-- ProteinA/  
|   |-- cell01/  
|   |   |-- loc_and_trc_file/  
|   |   |   |-- trcPALMTracer1.txt  
|   |   |   |-- trcPALMTracer2.txt  
|   |   |   |-- ...  
|   |-- cell02/  
|   |-- ...  
|-- ProteinB/  
|-- ...
```



Downloads > trapping-potential-map-spt-main > demo_data > ProteinA > cell01 > loc_and_trc_files				
Sort View ...				
Name	Date modified	Type	Size	
Today				
trcPALMTracer1	21-07-2026 11:29	Text Document	8,693 KB	
trcPALMTracer2	21-07-2026 11:29	Text Document	7,806 KB	
trcPALMTracer3	21-07-2026 11:29	Text Document	7,269 KB	
trcPALMTracer4	21-07-2026 11:29	Text Document	6,489 KB	
trcPALMTracer5	21-07-2026 11:29	Text Document	5,995 KB	

CONFIG CELL

Settings to check in the notebook before you run all cells

```
INPUT_FOLDERS = {  
    # r"D:\path\to\Protein\cell1\loc and trc files": r"D:\path\to\cell1\dendrite.rgn",  
    # r"D:\path\to\Protein\cell2\loc and trc files": None,  
    r"..demo_data\ProteinA\cell01\loc_and_trc_files": None  
}
```

When you want to process the entire image without any region boundary (ROI)

```
INPUT_FOLDERS = {  
    # r"D:\path\to\Protein\cell1\loc and trc files": r"D:\path\to\cell1\dendrite.rgn",  
    # r"D:\path\to\Protein\cell2\loc and trc files": None,  
    r"..demo_data\ProteinA\cell01\loc_and_trc_files": r"..demo_data\ProteinA\cell01\dendrite-region.rgn"  
}
```

When you want to process the image within a region boundary (ROI)

CONFIG CELL

Settings to check in the notebook before you run all cells

<pre># USER CONFIG # =====</pre>	
<pre># Mode: 1 = single-label, 2 = dual-label N_PROTEINS = 1</pre>	1 – single-label mode; one protein 2 – dual-label mode; two proteins
<pre># ----- Single-label input ----- # Keys : folder containing trcPALMTracer*.txt # Values : path to MetaMorph .rgn boundary file, or None for no boundary INPUT_FOLDERS = { # r"D:\path\to\Protein\cell1\loc and trc files": r"D:\path\to\cell1\dendrite.rgn", # r"D:\path\to\Protein\cell2\loc and trc files": None, r"..demo_data\ProteinA\cell01\loc_and_trc_files": r"..demo_data\ProteinA\cell01\dendrite-region.rgn" }</pre>	Single label only – dict mapping each trc folder to its .rgn boundary file (or None)
<pre># ----- Dual-label input (used only when N_PROTEINS == 2) ----- # Each value must be a 3-tuple (protein1_folder, protein2_folder, rgn_path_or_None). # Use None for the third entry when no boundary file is available. DUAL_INPUT_FOLDERS = { # "cell1": (# r"D:\Publications\Spring-Constant\Input_files_SC\Dual_color\cell1\PSD95\loc_and_trc_files", # r"D:\Publications\Spring-Constant\Input_files_SC\Dual_color\cell1\GluA2\loc_and_trc_files", # None, #) }</pre>	Dual-label only – dict mapping each cell name to a 3-tuple (protein1_folder, protein2_folder, rgn_path or None)
<pre># Protein display names (dual-label only) PROTEIN1_NAME = "ProteinA" # red panel PROTEIN2_NAME = "ProteinB" # blue panel</pre>	Display names for the red and blue channels in dual label mode (eg: "PSD95", "GluA2")
<pre># ----- Output ----- OUTPUT_ROOT = r"..Output_with_ROI\demo_data"</pre>	Path of the output files
<pre># ----- Camera calibration ----- NM_PER_PIXEL = 160.0 # nm per camera pixel</pre>	Camera calibration in nm per pixel
<pre># ----- Zoom (spatial binning) ----- # bin size = NM_PER_PIXEL / zoom. e.g., zoom=4 -> 40 nm bins at 160 nm/px camera. ZOOM_VALUES = [1, 2, 4, 8]</pre>	Pixel binning factor Bin_size = NM_PER_PIXEL / zoom

CONFIG CELL

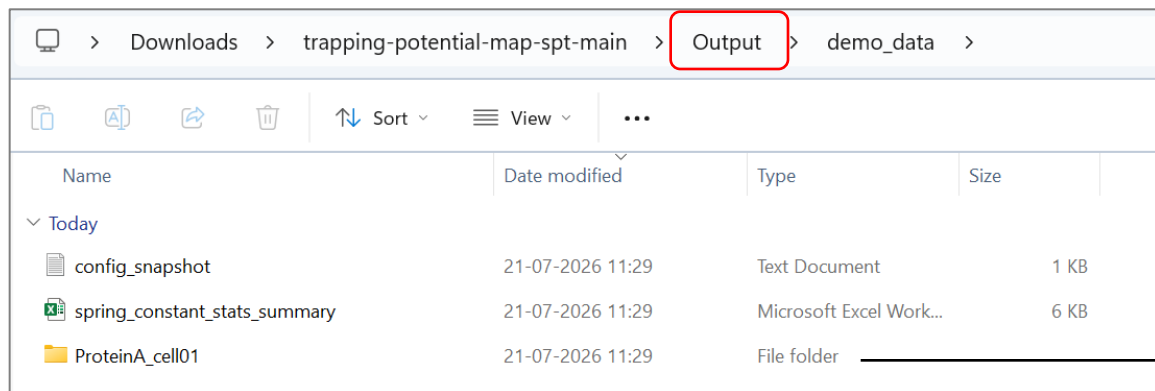
Settings to check in the notebook before you run all cells

<pre># ----- Track length filter ----- # Minimum localisations per track (summed across all frames). # Suggested: 6 = ensemble SPT default; 1-2 for sparse / single-trajectory data. MIN_POINTS = 6</pre>	Minimal localisations per track (total across all frames) Tracks shorter than this are discarded before pixel binning
<pre># ----- Colormap scale ----- # Log(k) range used for colour mapping. Leave as None to auto-derive on the # first single-label run, then paste the printed values back to lock the scale. # Dual-label mode REQUIRES explicit values. VMIN = None VMAX = None</pre>	log10(k) range for the color scale Leave both as None on the first single-label run to auto-derive from data; The printed values can then be pasted back to lock the scale across subsequent runs Note: Dual-label model requires both to be set explicitly
<pre># ----- Single-localisation handling ----- # True : segments with Rg^2 = 0 -> k = 1/(10 nm)^2 (bead / immobile control) # False : segments with Rg^2 = 0 -> NaN (rendered white, excluded from stats) USE_LOC_PRECISION_FLOOR = False</pre>	Controls what happens to a pixel whose only contributing track segment is a single localization False: single-localization pixels are set to NaN and rendered white on the map True: Rg ² is floored at the known localization precision squared (~ 20 nm) rather than 0. So single-localization pixel gets a finite k instead of blank
<pre># ----- Single-label discrete colormap (5 bins, low k -> high k) ----- BASE_COLORS = ["#FFE600", "#FF7518", "#8B00FF", "#0066FF", "#00CC00"]</pre>	Colors set for single-label and dual-label modes
<pre># ----- Dual-label gradients (6 bins each, white -> dark) ----- BASE_COLORS_RED = ["#FFFFFF", "#FFCCCC", "#FF8080", "#FF3333", "#CC0000", "#660000"] BASE_COLORS_BLUE = ["#FFFFFF", "#CCE5FF", "#80BFFF", "#3399FF", "#0055AA", "#001F4D"]</pre>	
<pre># ----- Dual-label difference thresholds ----- # d = Log(k1) - Log(k2): # d <= THRESH_WEAK -> black (co-confined) # THRESH_WEAK < d <= THRESH_STRONG -> light red / light blue # d > THRESH_STRONG -> dark red / dark blue THRESH_WEAK = 0.3 THRESH_STRONG = 1.0</pre>	Difference-map thresholds for dual-label mode only
<pre># ----- Boundary handling ----- MASK_TO_BOUNDARY = True # True: pixels outside .rgn polygon -> white BOUNDARY_LINEWIDTH = 0.3 # overlay linewidth</pre>	If set to True: pixels outside the .rgn polygon are set to white False: show the full field of view BOUNDARY_LINEWIDTH: line width of the ROI boundary overlay
<pre># ----- Parallelism ----- # None = all CPU cores. Set to 1 if you hit memory pressure. MAX_WORKERS = 1</pre>	None: all CPU cores, 1: serial (use if memory pressure occurs in parallel runs)
<pre># ----- Image generation ----- SAVE_IMAGES = True # False -> save .txt tables only (no TIFFs) DPI = 600 # 600/1200 for publication-quality TIFFs</pre>	False: skip all TIFF rendering, still saves .txt tables and .xlsx stats. Useful for fast numerical-only runs DPI: output image resolution

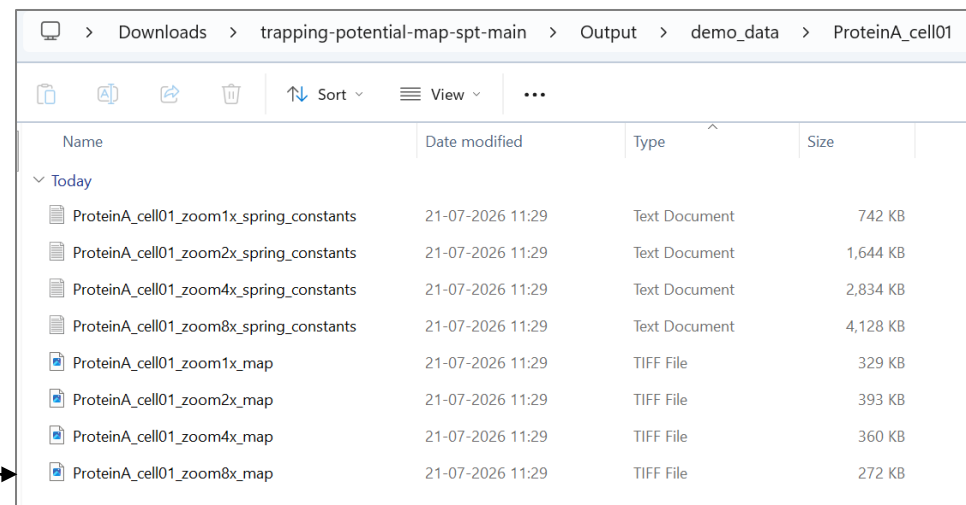
OUTPUT FILES

Everything lands under OUTPUT_ROOT, one sub-folder per cell

Without ROI

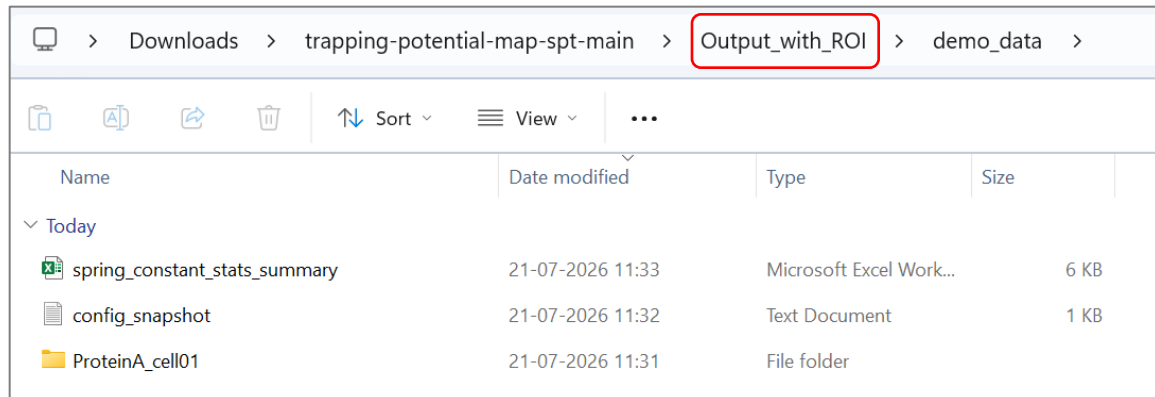


Name	Date modified	Type	Size
▼ Today			
config_snapshot	21-07-2026 11:29	Text Document	1 KB
spring_constant_stats_summary	21-07-2026 11:29	Microsoft Excel Work...	6 KB
ProteinA_cell01	21-07-2026 11:29	File folder	



Name	Date modified	Type	Size
▼ Today			
ProteinA_cell01_zoom1x_spring_constants	21-07-2026 11:29	Text Document	742 KB
ProteinA_cell01_zoom2x_spring_constants	21-07-2026 11:29	Text Document	1,644 KB
ProteinA_cell01_zoom4x_spring_constants	21-07-2026 11:29	Text Document	2,834 KB
ProteinA_cell01_zoom8x_spring_constants	21-07-2026 11:29	Text Document	4,128 KB
ProteinA_cell01_zoom1x_map	21-07-2026 11:29	TIFF File	329 KB
ProteinA_cell01_zoom2x_map	21-07-2026 11:29	TIFF File	393 KB
ProteinA_cell01_zoom4x_map	21-07-2026 11:29	TIFF File	360 KB
ProteinA_cell01_zoom8x_map	21-07-2026 11:29	TIFF File	272 KB

With ROI



Name	Date modified	Type	Size
▼ Today			
spring_constant_stats_summary	21-07-2026 11:33	Microsoft Excel Work...	6 KB
config_snapshot	21-07-2026 11:32	Text Document	1 KB
ProteinA_cell01	21-07-2026 11:31	File folder	

config_snapshot.txt – Every CONFIG value used in this run – your audit trail

spring_constant_stats_summary.xlsx – Descriptive stats median, IQR, mean +/- SEM) per zoom

<Protein>_<cell>_zoom<Z>x_spring_constants.txt – Per-pixel table: xid, yid, x/y (nm), Rg², k, log₁₀(k)

<Protein>_<cell>_zoom<Z>x_map.tiff – Discrete 5-colour log₁₀(k) map, one per zoom level

READING THE PER-PIXEL TABLE

Protein_A_cell01_zoom8x_spring_constants.txt

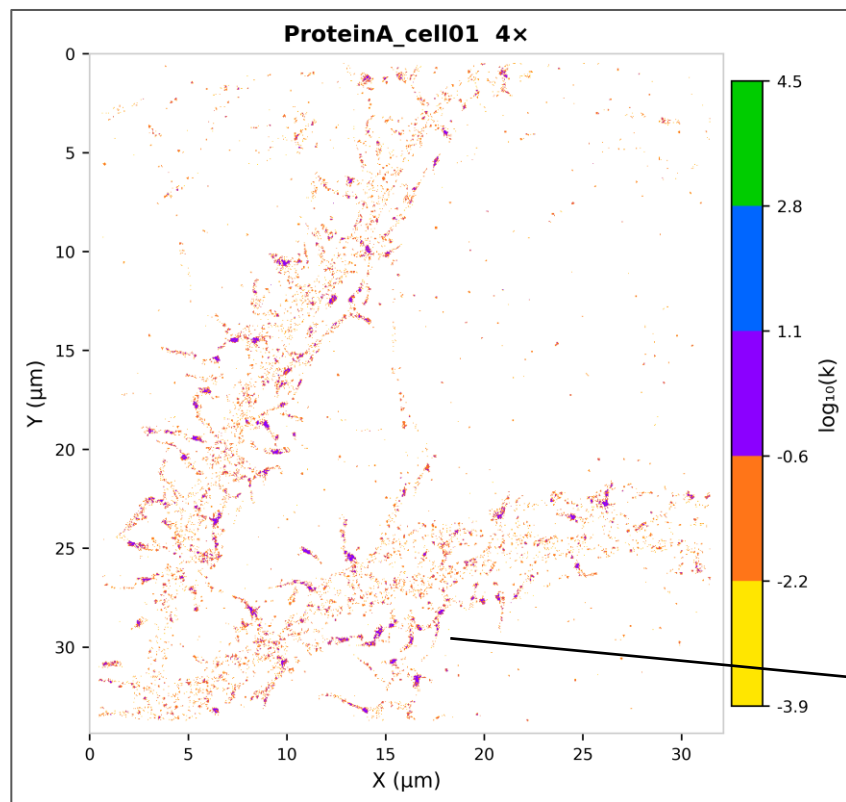
<xid>	<yid>	<x in nm>	<y in nm>	<Rg>	<Spring Const k>	<log10(k)>
24	1550	480.0	31000.0	0.25629955771356294	3.901684454397643	0.5912521432606439
24	1551	480.0	31020.0	0.5449345165060018	1.8350828764009597	0.26365568273025974
24	1581	480.0	31620.0			
24	1584	480.0	31680.0	1.11689125417142	0.8953423140034011	-0.048010890203912776
24	1585	480.0	31700.0	0.6219779060423456	1.6077741512765533	0.2062250420846041
24	1586	480.0	31720.0			

Blank Rg/ Spring Const k/ log10(k)
Pixel had only a single-localization segment ($Rg^2 = 0$) > Nan > white pixel on the map, excluded from statistics

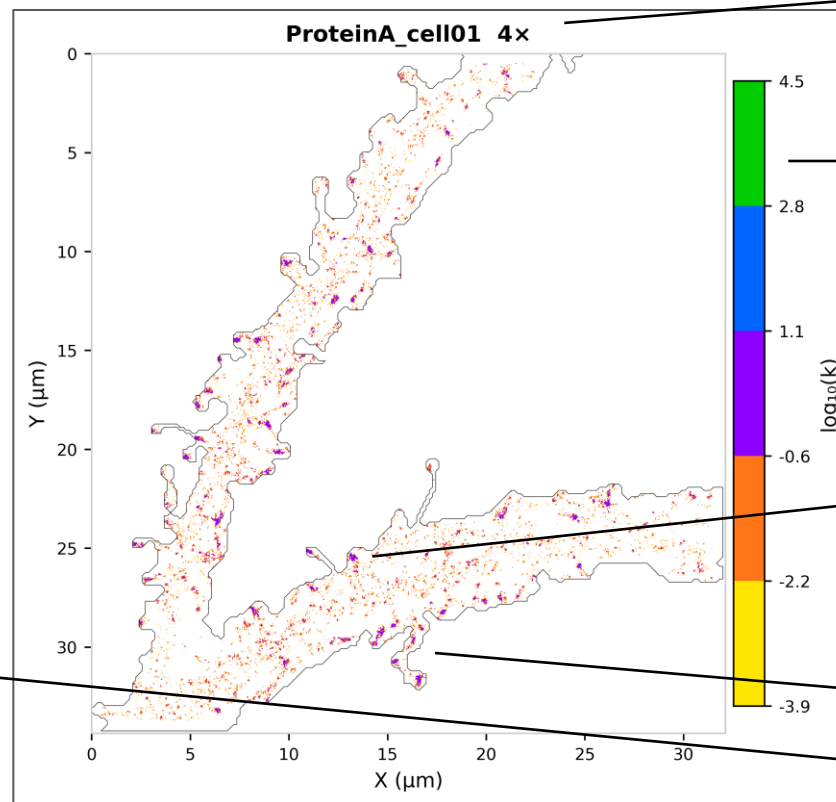
This is the value plotted
 $\log_{10}(k) = 0.2062\dots$ becomes one coloured pixel in the TIFF map

READING THE TIFF MAP

TIFF image without ROI



TIFF image with ROI



Label

Protein name, cell number, zoom

Log₁₀(k) colour scale

Yellow = Weak confinement (low k)

Green = Strong confinement (high k)

White = No data

Discrete-high k foci

Resolved once pixel size reaches zoom 4

Dendritic arbor shape

Emerges from ROI as well as from real tracks – confirms input data was read and binned correctly

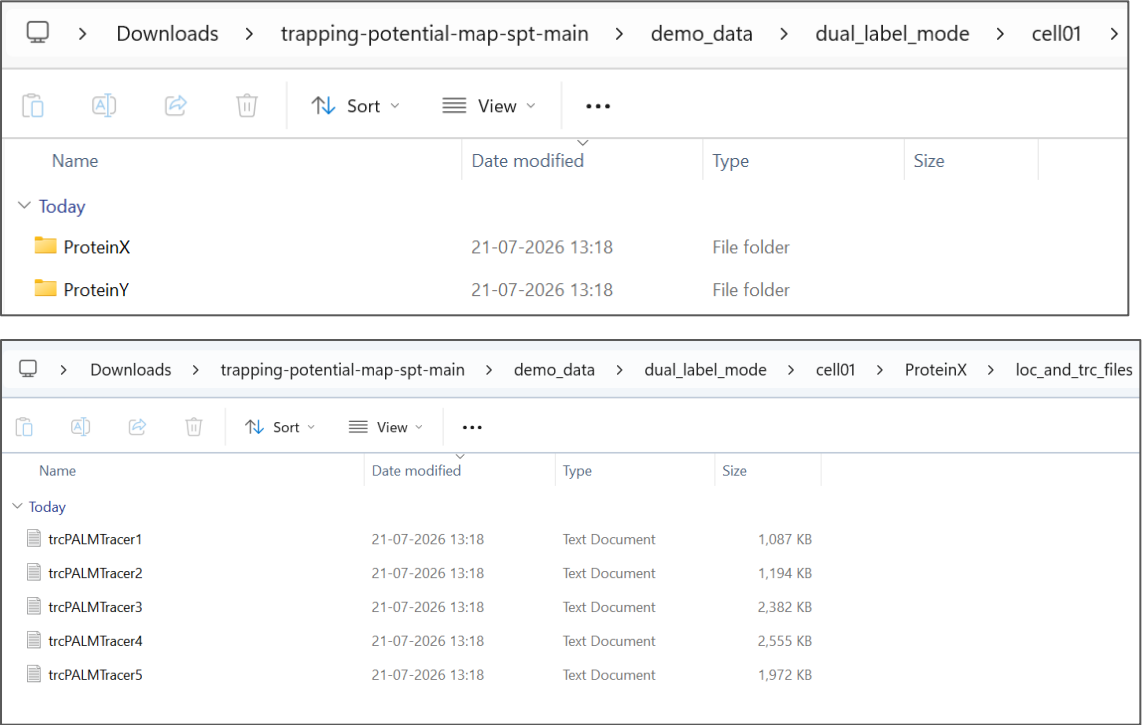
Dual-label mode

DUAL_INPUT_FOLDERS

DATA LAYOUT

Copy trc files out of the .PT folder into their own folder (loc_and_trc_files) following the hierarchy below

```
demo_data/  
|-- dual_label_mode/  
|   |-- cell01/  
|   |   |--ProteinX/  
|   |   |   |-- loc_and_trc_file/  
|   |   |   |   |-- trcPALMTracer1.txt  
|   |   |   |   |-- trcPALMTracer2.txt  
|   |   |   |   |-- ...  
|   |   |--ProteinY/  
|   |   |   |-- loc_and_trc_file/  
|   |   |   |   |-- trcPALMTracer1.txt  
|   |   |   |   |-- trcPALMTracer2.txt  
|   |   |   |   |-- ...  
|   |-- cell02/  
|   |-- ...
```



SETTING VMIN, VMAX VALUES

Setting VMIN, VMAX is necessary for dual-label mode.
If you are not sure of the VMIN, VMAX values then -
Before running dual-label mode, run all trc_files for both
proteins in INPUT_FOLDERS with N_PROTEINS = 1

```
# Mode: 1 = single-label, 2 = dual-label
N_PROTEINS = 1

# ----- Single-label input -----
# Keys   : folder containing trcPALMTracer*.txt
# Values : path to MetaMorph .rgn boundary file, or None for no boundary
INPUT_FOLDERS = {
    # r"D:\path\to\Protein\cell1\loc and trc files": r"D:\path\to\cell1\dendrite.rgn",
    # r"D:\path\to\Protein\cell2\loc and trc files": None,
    r"..demo_data\dual_label_mode\cell01\ProteinX\loc_and_trc_files": None,
    r"..demo_data\dual_label_mode\cell01\ProteinY\loc_and_trc_files": None
}
```

Use this VMIN, VMAX to lock the scale
for dual-label mode run

After a successful run you will get the global VMIN, VMAX
values that can be used for dual-label mode

12. Run

Edit the CONFIG cell above, then run this cell.

```
[122]: if N_PROTEINS == 1:
        main()
    elif N_PROTEINS == 2:
        main_dual()
    else:
        print(f"N_PROTEINS={N_PROTEINS!r} -- set to 1 or 2 in CONFIG")
```

-> Config snapshot saved: ..\Output_dual_label\demo_data\config_snapshot.txt

Auto-deriving colormap scale...

VMIN <- min log10(k) at zoom=1x across all cells

VMAX <- max log10(k) at zoom=8x across all cells

-> VMIN=-3.800, VMAX=5.745

Paste VMIN=-3.80 and VMAX=5.74 into CONFIG to lock the scale.

Colormap scale: log10(k) in [-3.80, 5.74]

Image generation: OFF (txt tables only)

=== ..demo_data\dual_label_mode\cell01\ProteinX\loc_and_trc_files ===

cell01_cellNA: 120,528 localisations, 5,090 unique tracks [frame 200x200 px]

zoom=1x (160nm/px): 2098 pixels log10k median=-2.90

zoom=2x (80nm/px): 5065 pixels log10k median=-1.82

zoom=4x (40nm/px): 12256 pixels log10k median=-0.57

zoom=8x (20nm/px): 26352 pixels log10k median=0.76

=== ..demo_data\dual_label_mode\cell01\ProteinY\loc_and_trc_files ===

cell01_cellNA: 179,480 localisations, 4,395 unique tracks [frame 200x200 px]

zoom=1x (160nm/px): 1575 pixels log10k median=-2.94

zoom=2x (80nm/px): 3865 pixels log10k median=-1.89

zoom=4x (40nm/px): 9562 pixels log10k median=-0.70

zoom=8x (20nm/px): 22426 pixels log10k median=0.58

-> Stats Excel saved: ..\Output_dual_label\demo_data\spring_constant_stats_summary.xlsx

All done.

CONFIG CELL

Settings to check in the notebook before you run all cells

```
DUAL_INPUT_FOLDERS = {
    # "cell1": (
    #     r"D:\Publications\Spring-Constant\Input_files_SC\Dual_color\cell1\PSD95\loc_and_trc_files",
    #     r"D:\Publications\Spring-Constant\Input_files_SC\Dual_color\cell1\GluA2\loc_and_trc_files",
    #     None,
    # )
    "cell01": (
        r"..demo_data\dual_label_mode\cell01\ProteinX\loc_and_trc_files",
        r"..demo_data\dual_label_mode\cell01\ProteinY\loc_and_trc_files",
        r"..demo_data\dual_label_mode\cell01\ProteinX_dendrite.rgn",
    )
}
```

→ Path of ProteinX

→ Path of ProteinY

→ Path of ROI boundary,
If no ROI boundary – set
this to None (as shown in
the comment)

Other Important CONFIG settings for dual-label mode

```
# Mode: 1 = single-label, 2 = dual-label
N_PROTEINS = 2
```

```
# ----- Colormap scale -----
# Log(k) range used for colour mapping. Leave as None to auto-derive on the
# first single-label run, then paste the printed values back to lock the scale.
# Dual-Label mode REQUIRES explicit values.
VMIN = -3.80
VMAX = 5.74
```

To set VMIN, VMAX see page 14

```
# ----- Dual-label difference thresholds -----
# d = Log(k1) - Log(k2):
# |d| <= THRESH_WEAK                -> black (co-confined)
# THRESH_WEAK < |d| <= THRESH_STRONG -> light red / light blue
# |d| > THRESH_STRONG                -> dark red / dark blue
THRESH_WEAK = 0.3
THRESH_STRONG = 1.0
```

```
# Protein display names (dual-label only)
PROTEIN1_NAME = "ProteinX" # red panel
PROTEIN2_NAME = "ProteinY" # blue panel
```

OUTPUT FILES

Everything lands under OUTPUT_ROOT, one sub-folder per cell

🖥️ > Downloads > trapping-potential-map-spt-main > Output_dual_label > demo_data >			
📁 📄 🔗 🗑️ ⬆️⬆️ Sort ▾ ≡ View ▾ ⋮			
Name	Date modified	Type	Size
▼ Today			
📄 config_snapshot	21-07-2026 13:50	Text Document	1 KB
📁 cell01	21-07-2026 13:50	File folder	

config_snapshot.txt – Every CONFIG value used in this run – your audit trail

<cell>_zoom<Z>x_dual.tiff – Gradient colour map for Protein X in red, ProteinY in blue and a difference map

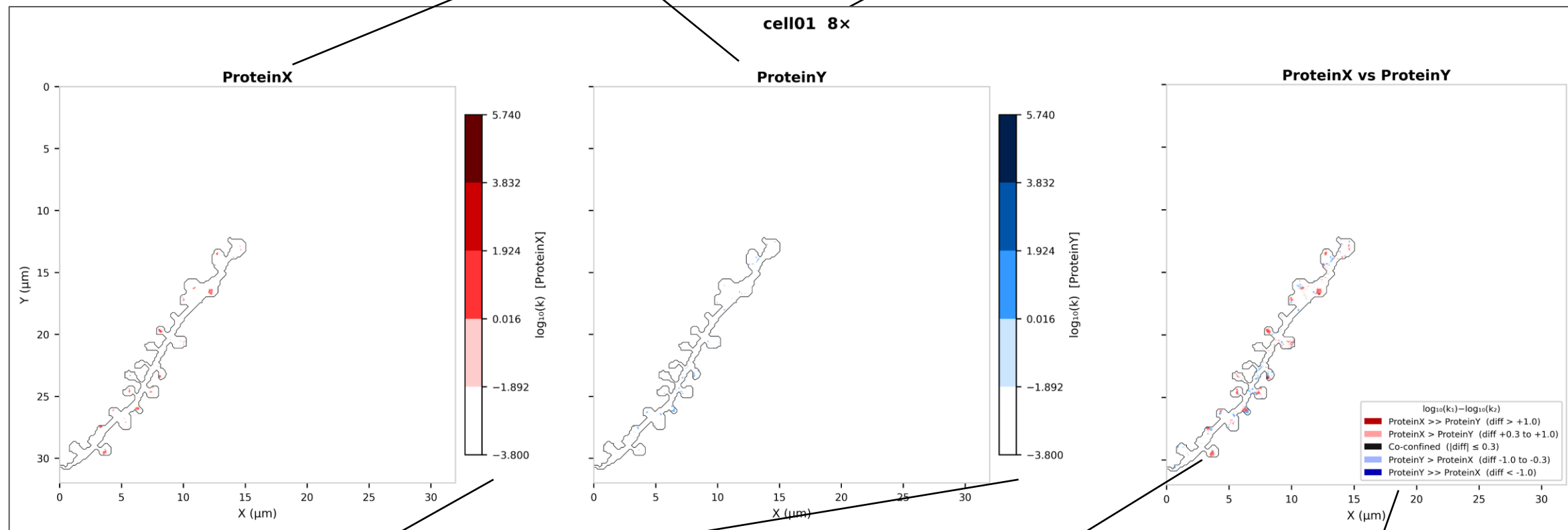
🖥️ > Downloads > trapping-potential-map-spt-main > Output_dual_label > demo_data > cell01			
📁 📄 🔗 🗑️ ⬆️⬆️ Sort ▾ ≡ View ▾ ⋮			
Name	Date modified	Type	Size
▼ Today			
📄 cell01_zoom8x_dual	21-07-2026 13:50	TIFF File	797 KB
📄 cell01_zoom4x_dual	21-07-2026 13:50	TIFF File	1,031 KB
📄 cell01_zoom2x_dual	21-07-2026 13:50	TIFF File	1,236 KB
📄 cell01_zoom1x_dual	21-07-2026 13:50	TIFF File	1,136 KB

Note: Dual-label mode produces only TIFF files. It is a display only feature and does not generate any .xlsx datasets

READING THE TIFF MAP

Label
Protein Name

Label
Cell number, zoom



log₁₀(k) colour scale

Light colour = Weak confinement (low k)
Dark colour = Strong confinement (high k)
White = No data

Dendritic arbor shape

Emerges from ROI as well as from real tracks – confirms input data was read and binned correctly

Difference-colour scale

Red = Higher localisation of ProteinX
Blue = Higher localization of ProteinY
Black = Co-confinement